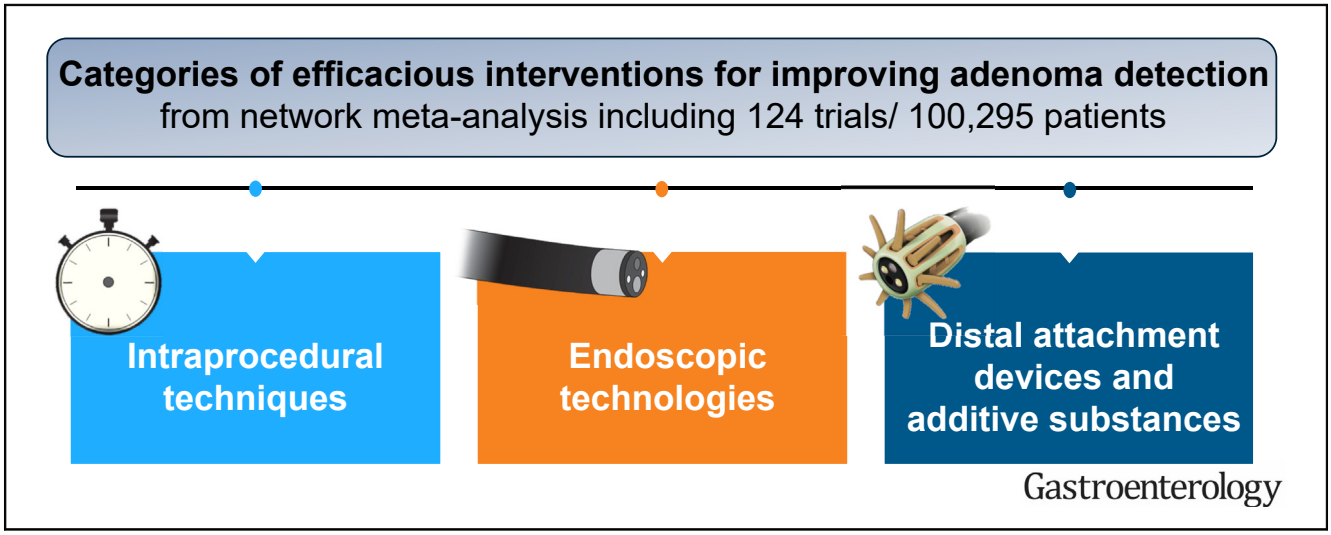
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PREVENTION AND EARLY DETECTION

Relative Efficacies of Interventions to Improve the Quality of Screening-Related Colonoscopy: A Systematic Review and Network Meta-Analysis of Randomized Controlled Trials

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BACKGROUND & AIMS: Significant variability exists in colonoscopy quality indicators, including adenoma detection rate (ADR). We synthesized evidence from randomized trials in a network meta-analysis on interventions to improve colonoscopy quality. **METHODS:** We included trials from database inceptions to September 25, 2023, of patients undergoing screening-related colonoscopy and presented efficacies of interventions within domains (periprocedural parameters, endoscopist-directed interventions, intraprocedural techniques, endoscopic technologies, distal attachment devices, and additive substances) compared to standard colonoscopy. The primary outcome was ADR. We used a Bayesian random-effects model using Markov-chain Monte Carlo simulation, with 10,000 burn-ins and 100,000 iterations. We calculated odds ratios with 95% credible intervals and present surface under the cumulative ranking (SUCRA) curves. **RESULTS:** We included 124 trials evaluating 37 interventions for the primary outcome. Nine interventions resulted in statistically significant improvements in ADR compared to standard colonoscopy (9-minute withdrawal time, dual observation, water exchange, i-SCAN [Pentax Ltd], linked color imaging, computer-aided detection, Endocuff [Olympus Corp], Endocuff Vision [Olympus Corp], and oral methylene blue). Dual observation (SUCRA, 0.84) and water exchange (SUCRA, 0.78) ranked highest among intraprocedural techniques; i-SCAN (SUCRA, 0.95), linked color imaging (SUCRA, 0.85), and computer-aided detection (SUCRA, 0.78) among endoscopic technologies; WingCap (A&A Medical Supply LLC) (SUCRA, 0.87) and Endocuff (SUCRA, 0.85) among distal attachment devices and oral methylene blue (SUCRA, 0.94) among additive substances. No interventions improved detection of advanced adenomas, and only narrow-band imaging improved detection of serrated lesions (odds ratio, 2.94; 95% credible interval, 1.46–6.25). **CONCLUSIONS:** Several interventions are effective in improving adenoma

detection and overall colonoscopy quality, many of which are cost-free. These results can inform endoscopists, unit managers, and endoscopy societies on relative efficacies.

Keywords: Colonoscopy; Screening; Adenoma; Colorectal Cancer.

Screening-related colonoscopy decreases colorectal cancer (CRC)–associated morbidity and mortality via detection and removal of premalignant adenomatous polyps.[1,2] Adenoma detection rate (ADR) is the most well-established colonoscopy performance indicator because of its inverse correlations with postcolonoscopy CRC as well as CRC-associated mortality.[3] Other established colonoscopy quality indicators include the cecal intubation rate and the rates of colonoscopy-associated adverse events.[4] The quality of screening-related colonoscopy as measured by performance indicators has been demonstrated to be highly variable among providers.[5–7]

There are numerous interventions and strategies available to boost provider-related performance and thereby enhance the quality of screening-related colonoscopy as measured and reported by quality indicators.[8,9] Examples include preprocedural strategies (eg, augmenting the quality of bowel cleansing), endoscopist-directed interventions (eg, directed audit and feedback[10] and educational courses[11]), intraprocedural techniques (eg, dynamic position changes), and artificial intelligence–based systems that focus on computer-aided detection (CADe).[12]

Despite a wealth of data on available interventions, practitioners may find it challenging to select intervention(s) to implement in their routine screening colonoscopy practice due to conflicting evidence on efficacy and effectiveness, considerations of cost, and overall acceptability and feasibility. Prior evidence syntheses on this topic have been limited by heterogeneity in study methodology, colonoscopy indications, patient populations, and comparator group definitions.[13–15] These limitations create challenges in interpreting the data and preclude the provision of meaningful guidance for practicing endoscopists regarding actionable colonoscopy quality improvement. To address this gap, we conducted a systematic review and network meta-analysis of randomized controlled trials comparing the impact of interventions, within predefined domains, on ADR and other established colonoscopy quality indicators.

WHAT YOU NEED TO KNOW

BACKGROUND AND CONTEXT

Adenoma detection rate (ADR) is an important colonoscopy quality metric associated with interendoscopist variability and can be improved on by using numerous interventions.

NEW FINDINGS

In this network-meta analysis of 124 trials, interventions improving ADR compared to standard colonoscopy included 9-minute withdrawal time, dual observation, water exchange, i-SCAN (Pentax Ltd), linked color imaging, computer-aided detection, Endocuff (Olympus Corp), Endocuff Vision (Olympus Corp), and oral methylene blue.

LIMITATIONS

Limitations include the exclusion of observational studies and abstracts, concerns about the generalizability of randomized trial populations, and lack of data on the impact of interventions on postcolonoscopy colorectal cancer.

CLINICAL RESEARCH RELEVANCE

Our findings highlight 9 interventions, several of which are readily available without any additional costs, that improve ADR in a screening-based colonoscopy population. Endoscopists can implement these interventions to improve colonoscopy quality for their patients. Studies are needed to evaluate cost-effectiveness.

BASIC RESEARCH RELEVANCE

The interventions that improved ADR accomplished this through different mechanisms, including enhancing mucosal exposure and reliably identifying adenomas. Future research is required to identify how these mechanisms interact with each other and whether they can be applied synergistically.

Methods

Overview

Network meta-analysis is a technique that can yield direct comparisons of interventions within studies and indirect comparisons of interventions across studies based on a common comparator.[16] Using our clinical question, examples of direct comparisons would be comparing a prolonged withdrawal time to standard colonoscopy in one trial and comparing a dedicated second examination of the right colon to standard colonoscopy in another trial. An indirect comparison that would then be enabled by network meta-analysis would be comparing prolonged withdrawal time to a second examination of the right colon, even though these 2 interventions may have never been directly compared. Network meta-analysis can often provide more accurate effect estimates than pairwise analyses[17] and is particularly useful when multiple interventions are available to improve a common desired outcome such as ADR because it enables investigators to rank these interventions in order of efficacy.

We conducted this systematic review and network meta-analysis according to the guidelines for Reporting of Systematic Reviews Incorporating Network Meta-analyses of Health Care Interventions (PRISMA-NMA).[18] Our protocol was registered a priori on PROSPERO (CRD42021291814) and published.[19] Institutional ethical approval was not required for this study given the lack of patient-specific data being collected.

Abbreviations used in this paper: **ADR, adenoma detection rate; CADe, computer-aided detection; CrI, credible interval; CRC, colorectal cancer; OR, odds ratio; SUCRA, surface under the cumulative ranking.**

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Search Strategy and Terms

We designed electronic search strategies for each interventional domain with a health research librarian (M.V.) of 7 databases (MEDLINE [Ovid], PubMed, Embase, CINAHL, Web of Science, Scopus, and Evidence-Based Medicine Reviews [EBMR]) from inception to February 1, 2022. We then updated the search twice: once on September 13, 2022, and then again on September 25, 2023 (the unabridged electronic search strategy is provided in Supplementary Table 1). We also manually cross-checked the reference lists of any relevant articles and/or prior evidence syntheses to ensure no trials were missed in our electronic search.

Eligibility Criteria

We aimed to achieve as bias-limited and uniform a cohort as possible that most closely reflected a true screening-related colonoscopy population. Therefore, we included only randomized controlled trials that met all of the following criteria: (1) patients were aged $\geq$18 years and undergoing screening-related colonoscopy; (2) interventions were within the prespecified domains[19]: preprocedural parameters, endoscopist-directed interventions, intraprocedural techniques, endoscopy technologies, distal attachment devices, and additive substances (with a full list of interventions by domain and associated descriptions provided in Supplementary Table 2); (3) the comparator(s) was/were either (a) "standard colonoscopy," defined as white light colonoscopy performed with a high-definition colonoscope or (b) any of the other interventions described within the prespecified domains; and (4) 1 or more of the primary or secondary outcomes was reported. In an attempt to create a truly homogeneous screening population, studies were considered to assess screening colonoscopy if they met all the following criteria[20]: (1) $\leq$15% of the study cohort fell outside the age range of 40 to 80 years, (2) $\leq$15% of the cohort were inpatients at the time of the procedure, and (3) $\leq$15% of the cohort were undergoing colonoscopy for non-screening-related indications (average-risk screening, positive stool-based test result, or surveillance given a personal or family history of colorectal polyps). We did not include balloon-assisted colonoscopy or similar interventions in our study given that these techniques generally require additional skills training.

Studies were excluded if any of the following criteria were met: (1) the comparator was unclear or did not represent standard colonoscopy or any of the interventions described; (2) the study included upper endoscopies and/or flexible sigmoidoscopies; (3) outcomes were assessed in trainees only; (4) the study included a high-risk population, such as those with hereditary polyposis syndromes; (5) the study was published before the year 2000 (given that it could contain data from non-high-definition colonoscopies); or (6) a conference abstract without a full-text publication was the only available published record.

Study Selection and Data Abstraction

We used Covidence (Melbourne, Australia) to import all search results and remove duplicate entries. Four reviewers (R.K., K.K., N.S.S., N.G.) performed screening, study selection, and data extraction in duplicate (ie, any 2 of the 4 reviewers performing a task completed it). Each reviewer was assigned an equal number of citations at each screening stage, with a vote of "both include" or "both exclude" by the 2 reviewers resulting in inclusion or exclusion, respectively. All discrepancies were resolved by a third vote by the senior study author (N.F.). Data were extracted in duplicate into standardized forms, with broad categories of data including study identification and design, endoscopist and patient characteristics, descriptions of the interventions and comparators, and outcomes. Data from all included studies were reviewed carefully to ensure that all included patients were unique and that there were no instances of multiple publications with overlapping cohorts of patients included in the final analyses.

Outcome Definitions

The primary outcome was ADR, defined as the proportion of colonoscopies during which 1 or more adenomas were detected. Secondary outcomes were polyp detection rate, ADR subtypes, adenoma per colonoscopy rate and subtypes, cecal intubation rate, withdrawal time, sedation requirements, adverse event rates, and overall rates of unplanned health encounters after colonoscopy. ADR subtypes included right-sided ADR (the proportion of colonoscopies in which at least 1 adenoma is found in the right colon), advanced ADR (the proportion of examinations with 1 or more adenomas of $\geq$10 mm in size or with high-grade dysplasia or a villous component) and sessile serrated ADR (the proportion of examinations with 1 or more sessile serrated adenomas). Similar subtypes were included for the adenoma per colonoscopy metric.

Risk of Bias

Two authors conducted risk of bias assessments for all studies included in the final review using the Cochrane Risk of Bias tool, version 2,[21] with any disagreements resolved by the senior author (N.F.). Using this tool, important study design and quality elements, such as blinding, missing data, and outcome measurement, can be evaluated and compared across studies.

Statistical Analysis

We first performed a Bayesian random-effects, pairwise meta-analysis to assess heterogeneity based on the I^2 statistic. In instances where substantial heterogeneity was observed, we planned to carry out metaregression on relevant covariates using random-effects models and by excluding studies.

For the network meta-analysis, we reported comparisons of interventions within the previously outlined domains: preprocedural parameters, endoscopist-directed interventions, intraprocedural techniques, endoscopy technologies, distal attachment devices, and additive substances. We reported the impact of specific interventions within each domain against standard colonoscopy for each available outcome. We used a Bayesian random-effects model using Markov-chain Monte Carlo simulation, with 10,000 burn-ins and 100,000 iterations. We calculated odds ratios (ORs) and corresponding 95% credible intervals (CrI) of network estimates with a consistency model. All ORs presented referred to the odds of the outcome occurring with the intervention relative to standard colonoscopy.

We present several graphical representations of our analyses, including one overall forest plot (a graph comparing effect sizes of all interventions included across all domains),

network diagrams (a plot summarizing the available direct comparisons of interventions within each domain), surface under the cumulative ranking (SUCRA) curves (graphs ranking probabilities of each intervention to be among the best interventions within each domain), and league tables (a matrix showing the relative effects between all interventions within each domain).

The transitivity assumption is a key element of appropriately designed network meta-analyses. If transitivity is present, investigators can assume that common comparators across studies are similar enough to enable indirect comparisons. We assessed the transitivity assumption of the network meta-analysis by examining the distribution of effect modifiers (sex and age of patients) across interventions to ensure no significant differences existed in these factors. We also carried out a test of inconsistency to determine whether there was statistical evidence of overall inconsistency. We assessed publication bias with a comparison-adjusted funnel plot and Egger test[22] and local incoherence by comparing direct and indirect estimates with a node-splitting method. We carried out analyses using the BUGSnet and gemtc R packages (R Foundation for Statistical Computing).[23]

Subgroup and Sensitivity Analyses

We planned to perform several sensitivity analyses to assess the following: low (<25%) vs high study attrition rates, results from studies published before 2015 vs in 2015 or later (to account for gradual overall improvements in colonoscopy quality over time[24]), results from North American vs European vs Asian studies, results from per-protocol vs intention-to-treat analyses, results from studies with clear and similar descriptions of comparator arms vs others, results from only those studies with even more restrictive screening-related criteria (<5% of study population meeting the described screening criteria rather than <15%), and results from all studies vs only those without high risks of bias according to the Risk of Bias 2 assessment tool. We also planned to perform subgroup analyses, where possible, to assess relevant patient subgroups (including patient age, sex, and comorbidities), indications for colonoscopy (including patients with positive fecal immunochemical test results vs surveillance vs initial screening vs others), bowel preparation quality, extent of trainee participation, sedation practices, colonoscopy completion rates, and endoscopist specialty and experience.

Certainty of the Evidence

We used the Grading of Recommendations Assessment, Development, and Evaluation (GRADE) approach and guidance on network meta-analysis to assess the certainty of evidence for the primary outcome of ADR.[16,25] We assessed risk of bias, indirectness, imprecision, publication bias, incoherence, and violation of transitivity for each comparison.

Results

From the 4962 unique records identified in our search strategy, we included 124 randomized trials with 100,295 unique patients in the network meta-analysis (a PRISMA study flow diagram is provided in Supplementary Figure 1, with a full list of citations for included studies provided in Supplementary Table 3). There were 22 trials in the periprocedural parameters domain, 23 in the intraprocedural techniques domain, 2 in the endoscopist-directed interventions domain, 42 in the endoscopic technologies domain, 23 in the distal attachment devices domain, 5 in the additive substances domain, and 7 that covered multiple domains (Table 1).

Among all trials, 47.7% of patients were women (Table 2). Mean and median ages were 59.0 and 60.4 years among the 107 trials that reported mean age and 17 trials that reported median age, respectively. There were 44 trials that recruited patients from Asia, 39 from Europe, 38 from North America, 2 from South America, and 1 from multiple continents. Furthermore, 52 trials were multicenter, and 72 were single-center. Reported colonoscopy indications were initial average-risk or family history–related screening for 64.2% of patients, follow-up of positive fecal immunochemical test or occult blood test result in 13.1%, surveillance due to prior personal history of polyps in 13.5%, and gastrointestinal symptoms in 9.2%. Bowel preparation was overall good/excellent. For the primary outcome of ADR, the Egger test did not show any significant evidence of publication bias (*P* values for 5 domains ranging from 0.18 to 0.92). Similarly, visual examination of funnel plots demonstrated no clear evidence of small study effects (with a full set of funnel plots provided in Supplementary File 1).

Adenoma Detection Rate

There were 37 individual interventions in the network meta-analysis (Figures 1 and 2) that reported on ADR. There were no interventions associated with improved ADR in the periprocedural parameters domain. In the intraprocedural techniques domain, 9-minute withdrawal time (1 trial, 733 participants; OR, 1.26; 95% CrI, 1.05–1.52), dual observation (3 trials, 1880 participants; OR, 1.15; 95% CrI, 1.03–1.29) and water exchange (5 trials, 2016 participants; OR, 1.11; 95% CrI, 1.02–1.21) significantly improved ADR compared to standard colonoscopy. In the endoscopic technologies domain, i-SCAN (Pentax Ltd) (4 trials, 1372 participants; OR, 1.33; 95% CrI, 1.11–1.63), linked color imaging (6 trials, 3568 participants; OR, 1.20; 95% CrI, 1.08–1.34), and CADe (11 trials, 11,538 participants; OR, 1.11; 95% CrI, 1.03–1.20) all improved ADR compared to standard colonoscopy. In the distal attachment devices domain, first-generation Endocuff (Olympus Corp) (5 trials, 2005 participants; OR, 1.30; 95% CrI, 1.14–1.51) and Endocuff Vision (Olympus Corp) (9 trials, 8798 participants; OR, 1.10; 95% CrI, 1.02–1.19) improved ADR compared to standard colonoscopy. In the additive substances domain, oral methylene blue (1 trial, 1002 participants; OR, 1.18; 95% CrI, 1.04–1.33) improved ADR compared to standard colonoscopy. All other interventions demonstrated no significant improvements in ADR compared to standard colonoscopy when studied within a stringent screening-related population (Table 3 and Figure 2). There were not enough studies to conduct network meta-analysis for the endoscopist-directed interventions domain.

A SUCRA plot for the intraprocedural techniques domain is provided in Figure 3, in which a 9-minute withdrawal time (SUCRA, 0.89), dual observation (SUCRA, 0.76), and

Table 1. Characteristics of Studies Included in the Network Meta-Analysis

Study	Location	Centers	Domain	Intervention	Indication	Number of patients	Outcomes included	Risk of bias
Adler (2009)	Germany	Multiple	Endoscopic technologies	Image-enhanced endoscopy: narrow-band imaging	Initial screen	1256	ADR, PDR, WT, CIR	High
Ahmad (2023)	United Kingdom	Single	Endoscopic technologies	CADe	FIT positive result, surveillance	608	ADR, ssADR, PDR, APC, ssAPC, WT, CIR	High
Alvarez-Gonzalez (2019)	Spain	Single	Periprocedural parameters	Precolonoscopy low-fiber diet	FIT positive result	276	ADR, rADR, PDR, CIR	High
Anderson (2019)	United States	Multiple	Intraprocedural techniques	Water immersion	Initial screen, symptoms, surveillance	121	WT	High
Aniwan (2021)	Thailand	Single	Multiple: endoscopic technologies and distal attachment devices	Image-enhanced endoscopy: linked color imaging, Endocuff Vision	Initial screen	1000	ADR, ssADR, rADR, aADR, APC	High
Aniwan (2023)	Thailand	Single	Multiple: endoscopic technologies and distal attachment devices	CADe, Endocuff Vision	Initial screen, FIT positive result	1245	ADR, rADR, aADR, APC, rAPC, aAPC, WT	High
Aslanian (2013)	United States	Single	Intraprocedural techniques	Dual observation	Initial screen, symptoms, surveillance	502	ADR, APC, WT	High
Avalos (2020)	United States	Single	Intraprocedural techniques	Segmental withdrawal	Initial screen	311	ADR, rADR, aADR, PDR, APC, WT, CIR	High
Bhattacharyya (2017)	United Kingdom	Single	Distal attachment devices	Endocuff Vision	FOBT positive result	531	ADR, aADR, PDR, APC, AE, WT, CIR	High
Biecker (2015)	Germany	Multiple	Distal attachment devices	First-generation Endocuff	Initial screen, symptoms, surveillance	498	ADR, PDR, APC, WT, CIR	High
Bogacki (2021)	Poland	Single	Endoscopic technologies	Wide-angle colonoscopy	Initial screen, symptoms, surveillance	421	ADR, aADR, PDR, WT	High

Table 1. Continued

Study	Location	Centers	Domain	Intervention	Indication	Number of patients	Outcomes included	Risk of bias
Bürger (2021)	Germany	Single	Endoscopic technologies	Image-enhanced endoscopy: narrow-band imaging	Initial screen, surveillance	473	ADR	High
Cadoni (2017)	Italy, Czech Republic	Multiple	Intraprocedural techniques	Water exchange, water immersion	Initial screen, FIT positive result, FOBT positive result	1224	ADR, ssADR, rADR, aADR, APC, rAPC, AE, WT, CIR	High
Calderwood (2011)	United States	Single	Periprocedural parameters	Patient education	Initial screen	969	PDR, APC, WT	High
Cheng (2019)	Taiwan	Single	Intraprocedural techniques	Inspection and polypectomy on insertion and withdrawal	Initial screen, surveillance	421	ADR, rADR, aADR, APC, WT	High
Cheng (2021)	Taiwan	Single	Intraprocedural techniques	Patient education	Initial screen, FIT positive result, surveillance	262	ADR, rADR, aADR, WT, CIR	High
Cheng (2023)	Taiwan	Single	Intraprocedural techniques	Water exchange, saline exchange	Initial screen, FIT positive result, surveillance	296	ADR, ssADR, rADR, APC, ssAPC, rAPC, WT, CIR	High
Choi (2020)	Korea	Single	Periprocedural parameters	Bowel preparation: bisacodyl and probiotics	Initial screen	385	ADR, PDR	Some concerns
Chung (2010)	Korea	Single	Endoscopic technologies	Image-enhanced endoscopy: flexible spectral imaging color enhancement	Initial screen	359	ADR, PDR, APC	High
Chung (2014)	Korea	Single	Endoscopic technologies	Image-enhanced endoscopy: narrow-band imaging, flexible spectral imaging color enhancement	Initial screen	1650	ADR, PDR, APC	High
Coghlan (2020)	Argentina	Single	Intraprocedural techniques	Segmental withdrawal	Initial screen	1149	ADR, WT	High
De Wijkerslooth (2011)	The Netherlands	Multiple	Distal attachment devices	Cap-assisted colonoscopy	Initial screen	1339	ADR, ssADR, rADR, aADR, APC, ssAPC, rAPC, aAPC, AE, WT, CIR	High

Table 1. Continued

Study	Location	Centers	Domain	Intervention	Indication	Number of patients	Outcomes included	Risk of bias
Desai (2022)	United States	Multiple	Distal attachment devices	Endocuff Vision, cap-assisted colonoscopy	Initial screen, surveillance	1142	ADR, ssADR, rADR, aADR, APC, AE, WT, CIR	High
Di Caro (2019)	United Kingdom	Multiple	Endoscopic technologies	High-definition colonoscopy	FOBT positive result	262	ADR, PDR, WT	High
dos Santos (2022)	Brazil	Single	Endoscopic technologies	Image-enhanced endoscopy: blue light imaging, image-enhanced endoscopy: linked color imaging	Initial screen	205	ADR	High
Eluri (2018)	United States	Multiple	Periprocedural parameters	Abdominal compression device	Initial screen, surveillance, symptoms	350	ADR, ssADR, rADR, PDR, APC, WT	Low
Etik (2019)	Turkey	Single	Periprocedural parameters	Precolonoscopy low-fiber diet, bowel preparation: sulfate-free polyethylene glycol	Initial screen	507	ADR, PDR	High
Facciorusso (2020)	Italy	Multiple	Endoscopist-directed interventions	Trainee involvement in colonoscopy	Initial screen, surveillance, symptoms	812	ADR, ssADR, rADR, aADR, PDR, APC	High
Floer (2014)	Germany	Multiple	Distal attachment devices	First-generation Endocuff	Initial screen, surveillance, symptoms	492	ADR, PDR, APC, AE, WT, CIR	High
Garborg (2012)	The Netherlands	Multiple	Endoscopic technologies	Ultrathin colonoscope	Initial screen	182	ADR, rADR, aADR, PDR, CIR	High
Garborg (2015)	Norway, Poland	Multiple	Intraprocedural techniques	Water exchange	Initial screen, surveillance	473	ADR, ssADR, rADR, aADR, PDR, APC, CIR	High
Glissen Brown (2021)	United States	Multiple	Endoscopic technologies	CADe	Initial screen, surveillance	223	ADR, PDR, APC, ssAPC, WT	High
Gonzalez-Fernandez (2017)	Mexico	Single	Distal attachment devices	First-generation Endocuff	Initial screen	337	ADR, rAPC	High
Gross (2011)	United States	Single	Endoscopic technologies	Image-enhanced endoscopy: narrow-band imaging	Initial screen, surveillance	96	ADR, PDR	High

Table 1. Continued

Study	Location	Centers	Domain	Intervention	Indication	Number of patients	Outcomes included	Risk of bias
Guo (2016)	China	Single	Intraprocedural techniques	Second forward view	Initial screen	360	ADR, rADR, aADR, PDR	High
Gweon (2020)	South Korea	Multiple	Intraprocedural techniques	Inspection and polypectomy on insertion and withdrawal	Initial screen, FOBT positive result, symptoms, surveillance	1142	ADR, APC, WT	High
Han (2021)	United States	Single	Additive substances	Peppermint/L-menthol	FOBT positive result, FIT positive result, surveillance	192	ADR	Low
Hasegawa (2021)	Japan	Single	Endoscopic technologies	Image-enhanced endoscopy: linked color imaging	Initial screen, symptoms, surveillance	700	ADR, aADR, PDR, APC, WT	High
Hassan (2016)	Italy	Multiple	Endoscopic technologies	Wide-angle colonoscopy	FIT positive result	658	ADR, ssADR, rADR, aADR	High
Hassan (2019)	Italy	Multiple	Distal attachment devices	EndoRings (STERIS Inc)	FIT positive result	634	ADR, ssADR, rADR, aADR, PDR, APC, WT, CIR	High
Hewett (2010)	United States	Single	Distal attachment devices	Cap-assisted colonoscopy	Initial screen, symptoms, surveillance	100	ADR, rADR	High
Hjelkrem (2011)	United States	Single	Periprocedural parameters	Bowel preparation: electrolyte-free polyethylene glycol	Initial screen	403	PDR	Some concerns
Hoffman (2010)	Germany	Single	Endoscopic technologies	Image-enhanced endoscopy: i-SCAN	Initial screen, FOBT positive result, surveillance	200	ADR, WT	High
Hoffman (2014)	Germany	Single	Endoscopic technologies	Image-enhanced endoscopy: i-SCAN	Initial screen, FOBT positive result, surveillance	80	ADR, PDR, WT	High
Hong (2011)	South Korea	Single	Endoscopic technologies	Image-enhanced endoscopy: i-SCAN	Initial screen	352	ADR, aADR, PDR, APC, WT	High
Hong SN (2022)	South Korea	Multiple	Periprocedural parameters	Bowel preparation: low-dose polyethylene glycol	Initial screen, symptoms, surveillance	346	ADR, rADR, PDR, APC	Some concerns
Hong SW (2022)	South Korea	Single	Distal attachment devices	WingCap	Initial screen, surveillance	528	ADR, aADR, PDR, APC, aAPC, AE, WT, CIR	High

Table 1. Continued

Study	Location	Centers	Domain	Intervention	Indication	Number of patients	Outcomes included	Risk of bias
Horimatsu (2015)	Japan	Multiple	Endoscopic technologies	Image-enhanced endoscopy: narrow-band imaging, wide-angle colonoscopy	Initial screen, surveillance, FOBT positive result	431	ADR, WT, CIR	High
Horiuchi (2010)	Japan	Single	Distal attachment devices	Cap-assisted colonoscopy, image-enhanced endoscopy: narrow-band imaging	Initial screen, surveillance, FOBT positive result	107	ADR, CIR	High
Ikematsu (2012)	Japan	Multiple	Endoscopic technologies	Image-enhanced endoscopy: narrow-band imaging	Initial screen, symptoms, surveillance	782	ADR, aADR, PDR	High
Inoue (2014)	Japan	Single	Additive substances	Peppermint/L-menthol	Initial screen, surveillance, FOBT positive result	226	ADR, WT, CIR	High
Inoue (2020)	Japan	Single	Intraprocedural techniques	Peppermint/L-menthol, carbon dioxide insufflation	Initial screen, surveillance, FOBT positive result	611	ADR, WT, CIR	High
Jaensch (2022)	Denmark	Single	Distal attachment devices	Endocuff Vision	Initial screen	1178	ADR, PDR, WT, CIR	High
Jeon (2015)	South Korea	Single	Periprocedural parameters	Bowel preparation: sulfate-based preparation	Initial screen	388	ADR, PDR, WT, CIR	Some concerns
Jung (2016)	Korea	Single	Preprocedural parameters	Bowel preparation: coffee with polyethylene glycol	Initial screen, surveillance	213	PDR, AE	Some concerns
Kahi (2010)	United States	Multiple	Additive substances	Chromoendoscopy: indigo carmine	Initial screen	660	ADR, PDR, APC, rAPC, aAPC	High
Kakol (2013)	Poland	Single	Endoscopic technologies	Image-enhanced endoscopy: narrow-band imaging	Initial screen, symptoms	253	ADR, rADR, WT	High
Kaltenbach (2008)	United States	Single	Endoscopic technologies	Image-enhanced endoscopy: narrow-band imaging	Initial screen, FOBT positive result, symptoms, surveillance	284	ADR, WT	High
Kamba (2021)	Japan	Multiple	Endoscopic technologies	CADe	Initial screen, symptoms, FOBT positive result, surveillance	355	ADR, ssADR, aADR, PDR, APC, AE, WT, CIR	High

Table 1. Continued

Study	Location	Centers	Domain	Intervention	Indication	Number of patients	Outcomes included	Risk of bias
Kidambi (2019)	United States	Single	Endoscopic technologies	Image-enhanced endoscopy: i-SCAN	Initial screen, surveillance, FIT positive result	740	ADR, ssADR, rADR, aADR, APC, AE, WT	High
Kim (2012)	South Korea	Single	Distal attachment devices	Cap-assisted colonoscopy	Initial screen	295	ADR, ssADR, CIR	High
Kim (2019)	South Korea	Single	Preprocedural parameters	Bowel preparation: simethicone with polyethylene glycol	Initial screen, symptoms, FOBT positive result	234	ADR, ssADR, rADR, PDR, AE, WT, CIR	High
Kim (2020a)	South Korea	Multiple	Multiple: endoscopic technologies and distal attachment devices	Cap-assisted colonoscopy, chromoendoscopy	Initial screen	1884	ADR, rADR, aADR, PDR, APC, rAPC, aAPC, AE, WT	High
Kim (2020b)	South Korea	Single	Intraprocedural techniques	Second forward view	Initial screen	640	ADR, aADR, PDR, AE, WT	High
Kondo (2007)	Japan	Single	Distal attachment devices	Cap-assisted colonoscopy	Initial screen, surveillance, symptoms	684	PDR	High
Kudo (2018)	Japan	Multiple	Endoscopic technologies	Wide-angle colonoscopy	Initial screen, surveillance, FOBT positive result, symptoms	319	ADR, ssADR, aADR, PDR, APC, AE, WT, CIR	High
Kudo (2021)	Japan	Single	Endoscopic technologies	Image-enhanced endoscopy: linked color imaging	Initial screen, surveillance, symptoms	302	ADR, ssADR, aADR, PDR, APC, AE, WT, CIR	High
Kumar (2017)	United States	Multiple	Intraprocedural techniques	Short withdrawal time	Initial screen, surveillance, symptoms	198	ADR, APC	High
Kushnir (2015)	United States	Multiple	Intraprocedural techniques	Second forward view	Initial screen, surveillance	850	ADR, rADR, aADR, APC, rAPC, AE, WT	High
Lee (2011)	South Korea	Multiple	Intraprocedural techniques	Dual observation	Initial screen	791	ADR, aADR, PDR, WT	High
Lee (2015)	South Korea	Single	Preprocedural parameters	Patient education	Initial screen	390	ADR, PDR, AE, WT, CIR	High
Leung (2009)	United States	Single	Intraprocedural techniques	Water immersion	Initial screen, surveillance	56	AE, CIR	High

Table 1. Continued

Study	Location	Centers	Domain	Intervention	Indication	Number of patients	Outcomes included	Risk of bias
Leung (2019)	United States	Single	Intraprocedural techniques	Water exchange, chromoendoscopy: indigo carmine	Initial screen, FIT positive result	480	ADR, ssADR, rADR, APC, AE, WT, CIR	High
Lovász (2022)	Hungary	Single	Endoscopic technologies	Image-enhanced endoscopy: linked color imaging	Initial screen	1278	ADR, PDR, APC, WT, CIR	High
Luo (2021)	China	Single	Endoscopic technologies	CADe	Initial screen, surveillance, symptoms	300	PDR, WT	High
Mangas-Sanjuan (2023)	Spain	Multiple	Endoscopic technologies	CADe	FIT positive result	3213	ADR, ssADR, aADR, PDR, APC, ssAPC, aAPC, WT	High
Marsano (2019)	United States	Single	Distal attachment devices	Cap-assisted colonoscopy, first-generation Endocuff	Initial screen, surveillance	126	ADR, ssADR, rADR, PDR, APC, AE, WT, CIR	High
Matsumura (2015)	Japan	Single	Periprocedural parameters	Precolonoscopy low-fiber diet	Initial screen or surveillance	91	ADR	Some concerns
Mohammed (2015)	Canada	Single	Periprocedural parameters	Bowel preparation: single-dose polyethylene glycol	Initial screen	237	AE	Some concerns
Morajevi (2019)	United States	Single	Periprocedural parameters	Bowel preparation: simethicone with polyethylene glycol	Initial screen	268	ADR, PDR, AE, WT	Some concerns
Nakashima (2023)	Japan	Single	Endoscopic technologies	CADe	Initial screen, FIT positive result, surveillance	415	ADR, aADR, APC, WT	High
Ngu (2018)	England	Multiple	Distal attachment devices	Endocuff Vision	Symptoms, surveillance program or with positive FOBT result as part of BCSP	1772	ADR, ssADR, PDR, APC, AE, WT	High
Nunez Rodriguez (2019)	Spain	Single	Endoscopic technologies	Wide-angle colonoscopy	FOBT positive result	238	ADR, PDR, APC	High
Nunez Rodriguez (2020)	Spain	Multiple	Intraprocedural techniques	Proximal colon retroflexion	FOBT positive result	648	rADR	High

Table 1. Continued

Study	Location	Centers	Domain	Intervention	Indication	Number of patients	Outcomes included	Risk of bias
Othman (2017)	United States	Single	Distal attachment devices	Cap-assisted colonoscopy	Initial screen, surveillance	440	ADR, aADR, PDR, APC, AE, WT, CIR	High
Ou (2014)	Canada	Single	Intraprocedural techniques	Position change	Initial screen, surveillance, symptoms	776	ADR, PDR, APC, AE, WT	High
Paggi (2009)	Italy	Single	Endoscopic technologies	Image-enhanced endoscopy: narrow-band imaging	Initial screen, FOBT positive result	211	ADR, aADR, WT	High
Paggi (2020)	Italy	Multiple	Endoscopic technologies	Image-enhanced endoscopy: linked color imaging	Initial screen, FIT positive result	649	ADR, ssADR, rADR, aADR, PDR, WT	High
Park (2016)	South Korea	Single	Periprocedural parameters	Patient education	Initial screen	502	PDR, WT	High
Patel (2017)	Canada	Single	Intraprocedural techniques	Carbon dioxide insufflation	Initial screen, surveillance	204	ADR, aADR, PDR, APC, WT, CIR	Low
Pisera (2019)	Poland	Multiple	Periprocedural parameters	Bowel preparation: sulfate-based preparation	Initial screen	13,500	ADR, ssADR, rADR, aADR, PDR	Some concerns
Radaelli (2017)	Italy	Multiple	Periprocedural parameters	Bowel preparation: single-dose polyethylene glycol	Initial screen, FIT positive result	690	ADR, rADR, aADR, AE, WT	Some concerns
Rastogi (2011)	Italy	Multiple	Endoscopic technologies	Image-enhanced endoscopy: narrow-band imaging	Initial screen, surveillance	630	ADR, rADR, aADR, WT	High
Rastogi (2012)	Italy	Single	Distal attachment devices	Cap-assisted colonoscopy	Initial screen, surveillance	420	ADR, rADR, aADR, WT	High
Repici (2019)	Italy	Multiple	Additive substances	Oral methylene blue	Initial screen, surveillance	1002	ADR, PDR	Low
Repici (2020)	Italy	Multiple	Endoscopic technologies	CADe	Initial screen, surveillance, FIT positive result	685	ADR, ssADR, rADR, aADR	High
Repici (2021)	Italy	Multiple	Periprocedural parameters	Bowel preparation: low-dose polyethylene glycol	Initial screen, surveillance, FIT positive result, symptoms	388	ADR, ssADR, rADR, PDR, WT, CIR	High

Table 1. Continued

Study	Location	Centers	Domain	Intervention	Indication	Number of patients	Outcomes included	Risk of bias
Rex (2007)	United States	Single	Endoscopic technologies	Image-enhanced endoscopy: narrow-band imaging	Initial screen, surveillance	434	ADR, WT, CIR	High
Rex (2016)	United States	Multiple	Endoscopic technologies	Image-enhanced endoscopy: narrow-band imaging	Initial screen, surveillance	800	ADR, aADR, PDR, APC, ssAPC, aAPC, WT	High
Rex (2018)	United States	Multiple	Multiple: endoscopic technologies and distal attachment devices	First-generation Endocuff, EndoRings, wide-angle colonoscopy	Initial screen, surveillance, symptoms	1188	ADR, PDR, APC, ssAPC	High
Rex (2019)	United States	Multiple	Distal attachment devices	AmplifEYE (Cantel/ Medivators), Endocuff Vision	Initial screen, surveillance, symptoms	592	ADR, ssADR	High
Rex (2020a)	United States	Multiple	Distal attachment devices	Endocuff Vision	Initial screen, surveillance	200	ADR, ssADR, APC, ssAPC	High
Rex (2020b)	United States	Multiple	Distal attachment devices	EndoRings	Initial screen	562	ADR, ssADR, PDR, APC, ssAPC, WT	High
Rondonotti (2022)	Italy	Multiple	Endoscopic technologies	CADe	Initial screen, FIT positive result	800	ADR, aADR, PDR	High
Senore (2014)	Italy	Multiple	Endoscopic technologies	Image-enhanced endoscopy: narrow-band imaging	Initial screen, FIT positive result	237	ADR, rADR, aADR, PDR	High
Shah (2019)	United States	Single	Additive substances	Peppermint/L-menthol	Initial screen	48	ADR, AE, WT	Low
Shan (2017)	China	Single	Endoscopic technologies	Image-enhanced endoscopy: i-SCAN	Initial screen, surveillance, symptoms	403	PDR, WT	High
Shan (2021)	China	Single	Intraprocedural techniques	Second forward view	Initial screen or surveillance	392	ADR, rADR, aADR, PDR, AE, WT	High
Shaukat (2022)	United States	Multiple	Endoscopic technologies	CADe	Initial screen, surveillance	1359	ADR, ssADR, PDR	High
Shavakhi (2021)	Iran	Single	Periprocedural parameters	Unsedated colonoscopy	Initial screen, surveillance, symptoms	246	PDR, AE, CIR	Low
Singh (2017)	Hong Kong, Thailand, Singapore, Australia	Multiple	Endoscopic technologies	Image-enhanced endoscopy: narrow-band imaging	Initial screen	1006	ADR, ssADR, WT	High

Table 1. Continued

Study	Location	Centers	Domain	Intervention	Indication	Number of patients	Outcomes included	Risk of bias
Sola-Vera (2019)	Spain	Single	Distal attachment devices	Endocuff Vision, cap-assisted colonoscopy	FOBT positive result	711	ADR, PDR, APC, AE, WT, CIR	High
Stolpman (2014)	United States	Single	Periprocedural parameters	Precolonoscopy low-fiber diet	Initial screen, surveillance	201	ADR, aADR, PDR, CIR	Some concerns
Tang (2022)	China	Multiple	Intraprocedural Techniques	Second forward view	Initial screen, surveillance	1011	ADR, aADR, PDR, WT	High
Walker (2021)	United States	Single	Periprocedural parameters	Patient education	Initial screen, surveillance	213	ADR, PDR	Some concerns
Wallace (2017)	United States	Multiple	Endoscopist-directed interventions	Endoscopist education	Initial screen, surveillance	11,765	ADR, aADR, PDR, WT	High
Wallace (2022)	United States	Multiple	Endoscopic technologies	CADe	Initial screen, surveillance	230	ADR, APC, WT, CIR	High
Wang (2018)	China	Single	Intraprocedural techniques	Dual observation	Initial screen, surveillance	587	ADR, rADR, PDR, WT, CIR	High
Wei (2023)	United States	Multiple	Endoscopic technologies	CADe	Initial screen, surveillance	769	ADR, ssADR, APC, ssAPC, rAPC, WT	High
Xu (2022)	China	Multiple	Endoscopic technologies	CADe	Initial screen, FIT positive result	3059	ADR, WT, CIR	High
Yang (2020)	China	Single	Periprocedural parameters	Bowel preparation: sulfate-based preparation	Initial screen, surveillance	586	PDR	Some concerns
Zhao (2023)	China	Multiple	Intraprocedural techniques	9-minute withdrawal time	Initial screen	733	ADR, ssADR, rADR, aADR, PDR, APC, ssAPC, aAPC, WT	High
Zimmerman-Fraedrich (2022)	Germany	Single	Distal attachment devices	Endocuff Vision	Initial screen	1416	ADR, aADR, PDR, WT, CIR	High
Zorzi (2022)	Italy	Multiple	Distal attachment devices	Endocuff Vision	Initial screen	1813	ADR, ssADR, rADR, aADR, PDR, AE, WT	High

aADR, advanced ADR; APC, adenomas per colonoscopy; aAPC, advanced adenomas per colonoscopy; AE, adverse event; BCSP, bowel cancer screening programme; CIR, cecal intubation rate; FIT, fecal immunochemical test; FOBT, fecal occult blood test; PDR, polyp detection rate; rADR, right ADR; rAPC, right-sided adenomas per colonoscopy; ssADR, sessile serrated ADR; ssAPC, sessile serrated adenomas per colonoscopy; WT, withdrawal time.

Table 2. Patient Characteristics From Studies Included in the Network Meta-Analysis

	Age, *y*, mean (SD) or median (IQR)		Male sex, %		Screening-related indication, %[a]	
Study	Control	Intervention	Control	Intervention	Control	Intervention
Adler (2009)	64.3 (7.1)	64.8 (6.5)	48.0	47.0	100.0	100.0
Ahmad (2023)	66.4 (5.4)	66.2 (5.4)	32.0	35.7	Not stated	Not stated
Alvarez-Gonzalez (2019)	60.1 (SD not stated)	59.9 (SD not stated)	48.0	56.0	100.0	100.0
Anderson (2019)	63.4 (7.5)	63.4 (8.4)	73.0	80.0	95.0	92.0
Aniwan (2021)	61.1 (6.3)	61.4 (6.7), 61.1 (6.6), 61.7 (6.8)[b]	33.2	35.6, 33.4, 37.2[b]	100.0	100.0
Aniwan (2023)	62 (6.8)	62.8 (6.8), 62.1 (6.9), 61.7 (7.0)[b]	42.9	42.6, 33.3, 42.9[b]	100	100
Aslanian (2013)	57.8 (9.4)	58.1 (9.6)	53.4	47.4	97.3	96.8
Avalos (2020)	57.9 (6.8)	57.8 (6.5)	34.8	34.6	100.0	100.0
Bhattacharyya (2017)	67 (64–71)	68 (63–70)	67.9	60.9	100.0	100.0
Biecker (2015)	68 (58–76)	65 (54–74)	51.8	48.2	Not stated	Not stated
Bogacki (2021)	Male, 62.2 (11.9); female, 62.3 (11.5)	Male, 66.0 (12.5); female, 65.3 (12.9)	40.2	43.0	Not stated	Not stated
Bürger (2021)	67.4 (10.7)	66.4 (9.9)	50.4	56.8	100.0	100.0
Cadoni (2017)	60.9 (6.2)	61.4 (6.2), 61.0 (6.3)[b]	55.1	54.9, 54.7[b]	100.0	100.0
Calderwood (2011)	57.1 (7.3)	57.1 (8.0)	42.0	42.0	100.0	100.0
Cheng (2019)	58.2 (8.5)	57.6 (8.5)	47.1	46.9	100.0	100.0
Cheng (2021)	57.3 (8.3)	56.7 (8.8)	51.1	46.6	100.0	100.0
Cheng (2023)	51.2 (10.0)	51.4 (10.7), 52 (10.9), 53 (11.5)[b]	47.1	57.3, 49.3, 45.4	100.0	100.0
Choi (2020)	49.0 (9.9)	47.9 (9.9)	Not stated	Not stated	100.0	100.0
Chung (2010)	50.3 (9.8)	51.0 (9.5)	64.6	69.1	100.0	100.0
Chung (2014)	51.5 (8.9)	51.4 (8.6), 51.3 (8.8)[b]	67.2	62.7, 61.8[b]	100.0	100.0
Coghlan (2020)	57.0 (6.0)	57.0 (6.0)	56.4	53.9	100.0	100.0
De Wijkerslooth (2011)	60 (55–66)	60 (55–66)	50.0	51.0	100.0	100.0
Desai (2022)	62.0 (8.9)	62.1 (8.7), 62.7 (9.3)[b]	81.3	80.5, 79.9[b]	100.0	100.0
Di Caro (2019)	66.3 (4.3)	66.1 (4.1)	65.1	63.2	100.0	100.0
dos Santos (2022)	Not stated	Not stated	44.3	34.9, 49.3[b]	100.0	100.0
Eluri (2018)	61.1 (8.1)	59.9 (8.7)	38.0	38.0	100.0	100.0
Etik (2019)	61 (54–66)	61 (55–68), 60 (54–68), 58 (54–66)[b]	34.5	35.4, 41.7, 36.3[b]	91.0	91.0
Facciorusso (2020)	61 (52–68)	59 (49–66)	53.2	55.9	100.0	100.0

Table 2. Continued

Study	Age, *y*, mean (SD) or median (IQR)		Male sex, %		Screening-related indication, %[a]	
	Control	Intervention	Control	Intervention	Control	Intervention
Floer (2014)	63 (53–73)	64 (54–73)	44.9	49.0	Not stated	Not stated
Garborg (2012)	59.4 (2.7)	59.7 (3.0)	57.0	57.4	100.0	100.0
Garborg (2015)	61.0 (3.7)	60 (4.5)	51.0	51.0	100.0	100.0
Glissen Brown (2021)	60.5 (8.5)	61.2 (9.8)	61.8	47.8	100.0	100.0
Gonzalez-Fernandez (2017)	62 (55–70)	60 (55–66)	24.0	29.0	100.0	100.0
Gross (2011)	64 (55–70)	62 (54–69)	85.0	84.0	Not stated	Not stated
Guo (2016)	54.9 (10.3)	55.0 (11.0)	56.0	64.4	100.0	100.0
Gweon (2020)	60.4 (7.7)	60.1 (8.9)	46.9	46.6	91.6	91.0
Han (2021)	59.3 (9.6)	59.4 (11.5)	50.0	57.8	93.3	91.2
Hasegawa (2021)	65.8 (12.3)	66.5 (10.8)	63.2	62.5	Not stated	Not stated
Hassan (2016)	60.4 (6.1)	60.1 (6.2)	51.5	50.6	100.0	100.0
Hassan (2019)	61.8 (6.9)	62.9 (7.2)	50.8	50.2	100.0	100.0
Hewett (2010)	62.9 (1.2)	61.0 (1.0)	60.0	54.0	92.0	96.0
Hjelkrem (2011)	54.2 (6.7)	54.1 (5.3), 53.8 (5.4), 55.4 (5.7)[b]	55.0	49, 57, 48[b]	100.0	100.0
Hoffman (2010)	64.0 (13.0)	62.0 (15.0)	50.0	56.0	100.0	100.0
Hoffman (2014)	Not stated	Not stated	Not stated	Not stated	Not stated	Not stated
Hong (2011)	48.9 (10.7)	50.4 (11.4), 49.6 (11.3)[b]	64.7	65.2, 65.3[b]	100.0	100.0
Hong SN (2022)	50.7 (13.5)	51.1 (13.3)	42.4	48.3	87.2	86.8
Hong SW (2022)	59.0 (13.1)	59.6 (12.0)	42.1	54.6	100.0	100.0
Horimatsu (2015)	64 (26–85)	66 (36–83), 66 (33–86), 64 (32–83)[b]	66.0	70, 68, 64[b]	100.0	100.0
Horiuchi (2010)	67.4 (12.0)	70.6 (9.5)	65.0	75.0	100.0	100.0
Ikematsu (2012)	63.3 (9.9)	63.2 (10.2)	70.0	69.0	93.9	94.6
Inoue (2014)	66 (27–90)	68 (33–87)	50.0	55.1	100.0	100.0
Inoue (2020)	61 (23–92)	60 (29–84), 60 (23–87), 62 (32–89)[b]	49.0	49.0, 46.7, 49.0[b]	100.0	100.0
Jaensch (2022)	63.3 (SD not stated)	63.0 (SD not stated)	57.6	61.2	100.0	100.0
Jeon (2015)	53.6 (13.8)	54.7 (13.1)	56.5	55.4	100.0	100.0
Jung (2016)	Not stated	Not stated	Not stated	Not stated	100.0	100.0
Kahi (2010)	58.0 (7.0)	59.0 (7.0)	63.0	60.0	100.0	100.0
Kakol (2013)	55.8 (SD not stated)	59.0 (SD not stated)	34.1	31.6	Not stated	Not stated
Kaltenbach (2008)	64.0 (10.0)	64.0 (10.0)	97.0	96.0	86.0	86.0
Kamba (2021)	61.4 (10.0)	61.6 (9.9)	76.8	76.4	100.0	100.0

Table 2. Continued

Study	Age, *y*, mean (SD) or median (IQR)		Male sex, %		Screening-related indication, %[a]	
	Control	Intervention	Control	Intervention	Control	Intervention
Kidambi (2019)	60.8 (7.6)	61.4 (7.5)	49.6	49.9	100.0	100.0
Kim (2012)	66.1 (14.8)	65.4 (15.3)	62.6	62.7	100.0	100.0
Kim (2019)	56.4 (14.3)	50.7 (13.7), 50.2 (14.8)[b]	57.7	54.4, 51.9[b]	100.0	100.0
Kim (2020a)	57.5 (7.7)	57.6 (7.9)	48.1	48.2	100.0	100.0
Kim (2020b)	53.7 (8.6)	53.5 (9.2)	63.8	66.5	100.0	100.0
Kondo (2007)	61.3 (13.0)	61.4 (13.1), 63.3 (12.7)[b]	57.9	62.0, 62.7[b]	Not stated	Not stated
Kudo (2018)	61.2 (9.7)	61.5 (9.2)	65.6	66.7	98.7	96.8
Kudo (2021)	62.5 (14.0)	63.2 (12.0)	49.0	53.0	100.0	100.0
Kumar (2017)	62.0 (8.0)	61.0 (9.0)	68.0	65.0	94.0	96.0
Kushnir (2015)	59.0 (8.3)	59.9 (9.3)	41.5	40.7	100.0	100.0
Lee (2011)	58.1 (7.3)	58.6 (7.4)	51.3	54.6	100.0	100.0
Lee (2015)	47.1 (11.8)	46.0 (12.2), 45.7 (12.4)[b]	53.5	62.7, 59.8[b]	Not stated	Not stated
Leung (2009)	59.0 (8.6)	60.0 (6.6)	92.9	89.3	100.0	100.0
Leung (2019)	60.8 (6.5)	60.5 (6.5), 61.1 (6.4)[b]	91.2	85.7, 88.1[b]	100.0	100.0
Lovász (2022)	52.0 (14.0)	52.0 (13.9)	49.2	49.5	100.0	100.0
Luo (2021)	41.3 (10.4)	41.3 (10.4)	51.0	51.0	100.0	100.0
Mangas-Sanjuan (2023)	60.6 (5.7)	60.7 (5.8)	53.2	53.7	100.0	100.0
Marsano (2019)	59.3 (7.4)	60.0 (6.7), 60.0 (6.8)[b]	54.8	57.1, 54.8[b]	100.0	100.0
Matsumura (2015)	62.7 (12.0)	64.5 (11.3)	66.7	63.0	Not stated	Not stated
Mohamed (2015)	55.5 (SD not stated)	53.5 (SD not stated)	64.0	53.0	Not stated	Not stated
Morajevi (2019)	56.3 (7.6)	57.0 (6.0)	37.2	33.8	Not stated	Not stated
Nakashima (2023)	60.6 (5.7)	54.9 (10.9)	69.7	73.9	100.0	100.0
Ngu (2018)	61.7 (11.7)	62.1 (11.1)	57.1	56.8	100.0	100.0
Nunez Rodriguez (2019)	62.4 (3.7)	63.1 (3.8)	53.8	63.0	100.0	100.0
Nunez Rodriguez (2020)	60.6 (5.8)	59.7 (5.6)	56.1	55.0	100.0	100.0
Othman (2017)	59.5 (7.3)	60.1 (7.3)	34.5	28.6	100.0	100.0
Ou (2014)	58.6 (9.3)	58.7 (9.3)	44.6	46.9	100.0	100.0
Paggi (2009)	61.1 (5.8)	59.6 (5.3)	56.5	54.3	100.0	100.0
Paggi (2020)	61.1 (5.8)	59.6 (5.3)	49.5	49.7	100.0	100.0
Park (2016)	47.3 (9.2)	49.2 (8.6)	66.3	62.8	100.0	100.0
Patel (2017)	61.0 (8.1)	60.6 (7.6)	50.9	48.0	100.0	100.0
Pisera (2019)	Not stated	Not stated	47.1	47.1	100.0	100.0

Table 2. Continued

Study	Age, *y*, mean (SD) or median (IQR)		Male sex, %		Screening-related indication, %[a]	
	Control	Intervention	Control	Intervention	Control	Intervention
Radaelli (2017)	59.6 (6.1)	59.7 (6.0)	51.9	48.7	100.0	100.0
Rastogi (2011)	59.7 (8.4)	62.0 (9.3), 60.2 (8.7)[b]	68.0	63.0, 65.0[b]	100.0	100.0
Rastogi (2012)	61.3 (7.3)	60.7 (7.2)	93.3	96.2	100.0	100.0
Repici (2019)	61.6 (6.8)	61.2 (6.8)	61.6	59.9	100.0	100.0
Repici (2020)	61.1 (10.6)	61.5 (9.7)	49.6	50.4	100.0	100.0
Repici (2021)	60.1 (12.0)	59.5 (11.3)	51.3	43.6	100.0	100.0
Rex (2007)	62.6 (8.7)	62.3 (8.4)	55.0	51.0	100.0	100.0
Rex (2016)	61.4 (SD not stated)	61.1 (SD not stated)	40.0	46.0	100.0	100.0
Rex (2018)	62.6 (8.3)	62.3 (8.2), 62.3 (7.9), 62.3 (8.7)[b]	53.0	53.0, 47.0, 51.0[b]	96.0	92.0, 93.0, 95.0[b]
Rex (2019)	62.1 (10.0)	63.2 (9.7)	47.0	43.0	94.0	92.0
Rex (2020a)	61.7 (9.9)	62.7 (8.9)	42.4	56.4	100.0	100.0
Rex (2020b)	61.9 (7.9)	61.0 (8.1)	49.0	54.0	100.0	100.0
Rondonotti (2022)	61 (55–67)	62 (56–68)	49.6	52.6	100.0	100.0
Senore (2014)	67.0 (2.7)	66.7 (2.7)	59.2	59.8	100.0	100.0
Shah (2019)	53.8 (3.9)	64.8 (4.9)	54.2	20.8	100.0	100.0
Shan (2017)	49.7 (13.1)	48.9 (13.4)	60.9	58.7	100.0	100.0
Shan (2021)	46.1 (13.2)	47.7 (12.6)	54.3	54.3	100.0	100.0
Shaukat (2022)	59.9 (8.8)	60.6 (8.9)	52.4	54.0	100.0	100.0
Shavakhi (2021)	50.5 (12.1)	49.2 (13.6)	41.9	44.3	Not stated	Not stated
Singh (2017)	58.3 (6.2)	58.4 (6.1)	42.4	46.4	100.0	100.0
Sola-Vera (2019)	65 (19–92)	64 (18–92)	52.7	49.4	100.0	100.0
Stolpman (2014)	60.0 (8.9)	60.0 (8.4)	52.5	61.0	100.0	100.0
Tang (2022)	59.8 (8.5)	59.9 (8.4)	54.2	53.8	100.0	100.0
Walker (2021)	57.5 (1.1)	57.6 (1.2)	37.3	43.2	100.0	100.0
Wallace (2017)	59 (52–65)	59 (53–66)	44.0	49.0	100.0	100.0
Wallace (2022)	64.6 (8.1)	63.0 (8.2)	67.5	69.0	100.0	100.0
Wang (2018)	53.4 (8.1)	53.2 (8.5)	44.7	48.6	100.0	100.0
Wei (2023)	57.6 (7.9)	58.0 (7.9)	51.2	50.8	100.0	100.0
Xu (2022)	57.0 (7.4)	57.5 (7.6)	47.3	46.5	100.0	100.0
Yang (2020)	46.4 (9.8)	43.3 (10.5)	55.3	53.2	Not stated	Not stated
Zhao (2023)	53.6 (8.1)	54.1 (8.5)	47.1	49.7	100.0	100.0
Zimmerman-Fraedrich (2022)	61.1 (6.0)	61.1 (6.1)	47.1	49.3	100.0	100.0
Zorzi (2022)	60.1 (6.7)	60.2 (6.5)	53.8	53.7	100.0	100.0

IQR, interquartile range; SD, standard deviation.
[a]Screening-related indication includes initial screen, patients with positive result on FIT or FOBT, and surveillance for personal or family history of polyps or cancer.
[b]Multiple experimental arms.



Figure 1. Network diagrams for primary outcome (ADR) by domain. 2nd fwd, second forward view; abd comp dev, abdominal compression device; BLI, blue light imaging; CADe, computer-aided detection; Dual obs, dual observation; EC 1G, first-generation Endocuff; ECV, Endocuff vision; exch, exchange; FICE, flexible spectral imaging color enhancement; imm, immersion; Insp on ins, inspection on insertion; LCI, linked color imaging; LD PEG, low-dose polyethylene glycol; LFD, low-fiber diet; NBI, narrow-band imaging; Pat edu, patient education; PEG, polyethylene glycol; Pos, prescribed position change; SD-PEG, single-dose polyethylene glycol; SF-PEG, sulfate-free polyethylene glycol; Std CLN, standard colonoscopy; WA, wide angle; WT, withdrawal time.

saline exchange (SUCRA, 0.72) ranked highest among all interventions. In general, the higher the SUCRA value (ie, the closer it is to 1.00), the higher the likelihood of an intervention being ranked the highest in its domain. Therefore, based on this plot, 9-minute withdrawals are most likely to result in increased adenoma detection, followed by dual observation and then by saline exchange. In SUCRA analyses for other domains, low-fiber diet plus sulfate-free polyethylene glycol (SUCRA, 0.81) was ranked highest in the periprocedural parameters domain; i-SCAN (SUCRA, 0.96), linked color imaging (SUCRA, 0.84), and CADe (SUCRA, 0.68) in the endoscopic technologies domain; WingCap (A&A Medical Supply LLC) (SUCRA, 0.87) and first-generation Endocuff (SUCRA, 0.85) in the distal attachment devices domain, and oral methylene blue (SUCRA, 0.94) in the additive substances domain (Table 3). Remaining comparative evaluations of individual interventions through SUCRA plots and league tables according to prespecified domains are available in the Supplementary File 2 (SUCRA plots) and Supplementary File 3 (league tables).

Secondary Outcomes

All secondary outcomes are summarized in Table 4.

Adenoma detection rate subtypes. We conducted network meta-analysis for 3 domains (intraprocedural techniques, endoscopic technologies, distal attachment devices) for advanced ADR, 4 domains (periprocedural parameters, intraprocedural techniques, endoscopic technologies, distal attachment devices) for right-sided ADR, and 2 domains (endoscopic technologies, distal attachment devices) for serrated ADR. Water exchange (5 randomized trials, 2016 participants; OR, 1.15; 95% CrI, 1.02–1.31) in the intraprocedural techniques domain resulted in improved right-sided ADR, and narrow-band imaging (2 trials, 1426 participants; OR, 2.94; 95% CrI, 1.46–6.25) in the endoscopic technologies domain resulted in improved serrated ADR compared to standard colonoscopy. No interventions were associated with improved advanced ADR.

Polyp detection rate. We conducted network meta-analysis for 4 domains (periprocedural parameters, intraprocedural techniques, endoscopic technologies, distal

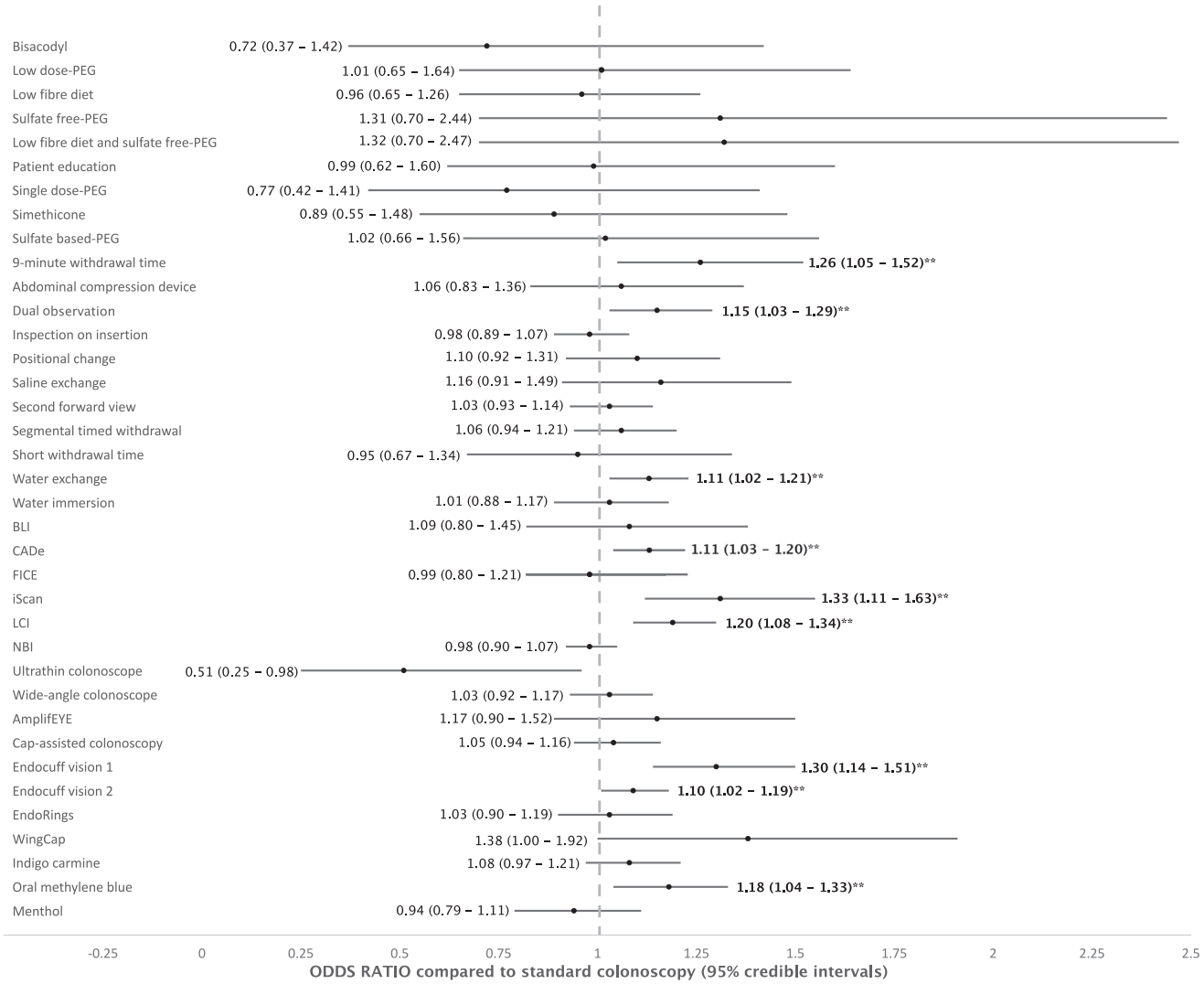


Figure 2. Forest plot with ORs and 95% CrI from network meta-analysis of the primary outcome (ADR). All comparisons to standard colonoscopy. **Interventions with statistically significant effects on ADR marked in bold and appear on the right of entries. BLI, blue light imaging; FICE, flexible spectral imaging color enhancement; LCI, linked color imaging; NBI, narrow-band imaging; PEG, polyethylene glycol.

attachment devices) for polyp detection rate. Use of linked color imaging (4 trials, 2929 participants; OR, 1.14; 95% CrI, 1.06–1.23) and CADe (6 trials, 6058 participants; OR, 1.06; 95% CrI, 1.02–1.12) in the endoscopic technologies domain and first-generation Endocuff (3 trials, 1584 participants; OR, 1.21; 95% CrI, 1.06–1.42) in the distal attachment devices domain all resulted in improved polyp detection rates. All other comparisons showed no significant improvement in polyp detection rates.

Adenoma per colonoscopy rate and subtypes. We conducted network meta-analysis for 3 domains for adenoma per colonoscopy rate (intraprocedural techniques, endoscopic technologies, distal attachment devices) and 2 domains for serrated adenoma per colonoscopy rate (intraprocedural techniques, distal attachment devices). Use of linked color imaging (2 trials, 1978 participants; mean difference, 0.19; 95% CrI, 0.09–0.30) and CADe (6 trials, 5583 participants; mean difference, 0.16; 95% CrI, 0.06–0.26) in the endoscopic technologies domain resulted in improved adenoma per colonoscopy rate. No interventions resulted in improved serrated adenoma per colonoscopy rate.

Cecal intubation rate and withdrawal time. We conducted network meta-analysis for 4 domains for cecal intubation rate and withdrawal time (periprocedural parameters, intraprocedural techniques, endoscopic technologies, distal attachment devices). No interventions resulted in significant improvements in cecal intubation rate. The use of narrow-band imaging in the endoscopic technologies domain resulted in increased withdrawal time (5 trials, 2605 patients; mean difference, 0.61 minutes; 95% CrI, 0.22–1.01 minutes).

Table 3. Summary of Findings for the Primary Outcome (ADR)

Intervention	Number of studies	Number of participants	Relative effect, OR (95% CrI)	Total anticipated effect, ADR			Certainty of evidence	SUCRA	Ranking
				Without intervention, %	With intervention, %	Difference, % (95% CrI)			
Periprocedural parameters									
Bisacodyl	1	385	0.72 (0.37 to 1.42)	25.8	19.0	–6.8 (–15.1 to 1.5)	Moderate (due to serious imprecision)	0.19	9
Low-dose PEG	2	734	1.01 (0.65 to 1.64)	30.5	31.2	0.7 (–8.0 to 9.4)	Very low (due to serious risk of bias, serious imprecision, and serious inconsistency)	0.54	4
Low-fiber diet	4	819	0.96 (0.65 to 1.26)	49.8	49.1	–0.7 (–13.4 to 12.0)	Very low (due to serious risk of bias, serious imprecision, and serious inconsistency)	0.46	6
Sulfate-free PEG	1	251	1.31 (0.70 to 2.44)	29.8	28.3	–1.5 (–12.7 to 9.7)	Low (due to serious risk of bias and serious imprecision)	0.80	2
Low-fiber diet and sulfate-free PEG	1	253	1.32 (0.70 to 2.47)	29.8	28.6	–1.2 (–12.4 to 10.1)	Low (due to serious risk of bias and serious imprecision)	0.81	1
Patient education	2	603	0.99 (0.62 to 1.60)	32.7	33.1	0.4 (–8.3 to 9.2)	Low (due to serious risk of bias and serious imprecision)	0.51	5
Single-dose PEG	1	690	0.77 (0.42 to 1.41)	53.0	40.8	–12.2 (–19.6 to 4.8)	Moderate (due to serious imprecision)	0.21	8
Simethicone	2	502	0.89 (0.55 to 1.48)	29.7	27.1	–2.6 (–10.6 to 5.3)	Moderate (due to serious imprecision)	0.37	7
Sulfate-based preparation	2	13,888	1.02 (0.66 to 1.56)	35.1	35.9	0.8 (–0.7 to 2.3)	Moderate (due to serious imprecision)	0.57	3

Table 3. Continued

Intervention	Number of studies	Number of participants	Relative effect, OR (95% CrI)	Total anticipated effect, ADR			Certainty of evidence	SUCRA	Ranking
				Without intervention, %	With intervention, %	Difference, % (95% CrI)			
Intraprocedural techniques									
9-minute withdrawal time	1	733	1.26 (1.05 to 1.52)	33.5	42.3	8.8 (1.8 to 15.8)	Low (due to very serious risk of bias)	0.89	1
Abdominal compression device	1	350	1.06 (0.83 to 1.36)	40.0	42.9	2.9 (–7.5 to 13.2)	Moderate (due to serious imprecision)	0.49	6
Dual observation	3	1880	1.15 (1.03 to 1.29)	35.1	41.3	6.2 (1.9 to 10.5)	Low (due to very serious risk of bias)	0.76	2
Inspection on insertion	2	1563	0.98 (0.89 to 1.07)	55.7	55.1	–0.6 (–5.7 to 4.4)	Low (due to very serious risk of bias)	0.19	11
Prescribed positional change	1	776	1.10 (0.92 to 1.31)	37.9	41.8	3.9 (–3.0 to 10.7)	Very low (due to very serious risk of bias and serious imprecision)	0.59	5
Saline exchange	1	221	1.16 (0.91 to 1.49)	50	50.3	0.3 (–13.8 to 14.5)	Very low (due to very serious risk of bias and serious imprecision)	0.72	3
Second forward view	4	2242	1.03 (0.93 to 1.14)	36.6	38.1	1.5 (–2.5 to 5.5)	Low (due to very serious risk of bias)	0.38	8
Segmental timed withdrawal	2	1460	1.06 (0.94 to 1.21)	38.7	41.3	2.6 (–2.4 to 7.6)	Low (due to very serious risk of bias)	0.49	6
Short withdrawal time	1	198	0.95 (0.67 to 1.34)	40.6	39.2	–1.4 (–15.1 to 12.2)	Very low (due to very serious risk of bias and serious imprecision)	0.26	10
Water exchange	5	2016	1.11 (1.02 to 1.21)	48.5	53.5	5.0 (0.1 to 9.8)	Low (due to very serious risk of bias)	0.64	4
Water immersion	1	816	1.01 (0.88 to 1.17)	40.4	43.3	2.9 (–3.8 to 9.7)	Low (due to very serious risk of bias)	0.33	9

Table 3. Continued

Intervention	Number of studies	Number of participants	Relative effect, OR (95% CrI)	Total anticipated effect, ADR			Certainty of evidence	SUCRA	Ranking
				Without intervention, %	With intervention, %	Difference, % (95% CrI)			
Endoscopic technologies									
Blue light imaging	1	136	1.09 (0.80 to 1.45)	52.9	62.2	9.3 (–7.3 to 25.8)	Very low (due to very serious risk of bias and serious imprecision)	0.57	4
CADe	11	11,538	1.11 (1.03 to 1.20)	47.9	53.5	5.6 (1.2 to 9.9)	Low (due to very serious risk of bias)	0.68	3
Flexible spectral imaging color enhancement	2	1459	0.99 (0.80 to 1.21)	26.5	25.9	–0.6 (–5.1 to 3.9)	Low (due to very serious risk of bias)	0.35	6
i-SCAN	4	1372	1.33 (1.11 to 1.63)	23.8	35.4	11.6 (2.2 to 21.0)	Very low (due to very serious risk of bias and serious inconsistency)	0.96	1
Linked color imaging	6	3568	1.20 (1.08 to 1.34)	46.7	55.0	8.3 (5.1 to 11.4)	Low (due to very serious risk of bias)	0.84	2
Narrow-band imaging	12	7099	0.98 (0.90 to 1.07)	42.9	42.1	–0.8 (–3.0 to 1.3)	Low (due to very serious risk of bias)	0.28	7
Ultrathin colonoscope	1	182	0.51 (0.25 to 0.98)	23.9	12.8	–11.1 (–22.3 to 0.1)	Low (due to very serious risk of bias)	0.02	8
Wide-angle colonoscope	5	2230	1.03 (0.92 to 1.17)	48.7	50.5	1.8 (–3.8 to 7.4)	Low (due to very serious risk of bias)	0.46	5
Distal attachment devices									
AmplifEYE	1	592	1.17 (0.90 to 1.52)	N/A	—	—	Very low (due to very serious risk of bias and serious imprecision)	0.54	3
Cap-assisted colonoscopy	6	3146	1.05 (0.94 to 1.16)	48.3	49.2	0.9 (–4.3 to 6.0)	Low (due to very serious risk of bias)	0.26	5

Table 3. Continued

Intervention	Number of studies	Number of participants	Relative effect, OR (95% CrI)	Total anticipated effect, ADR			Certainty of evidence	SUCRA	Ranking
				Without intervention, %	With intervention, %	Difference, % (95% CrI)			
First-generation Endocuff	5	2005	1.30 (1.14 to 1.50)	34.0	45.2	11.2 (6.9 to 15.4)	Very low (due to very serious risk of bias and serious inconsistency)	0.77	2
Endocuff Vision	9	8798	1.10 (1.02 to 1.19)	49.4	54.8	5.4 (2.1 to 8.8)	Very low (due to very serious risk of bias and serious inconsistency)	0.43	4
EndoRings	3	1786	1.03 (0.90 to 1.19)	51.7	51.9	0.2 (–5.6 to 6.0)	Low (due to very serious risk of bias)	0.23	6
WingCap	1	528	1.38 (1.00 to 1.91)	26.6	37.1	10.5 (2.6 to 18.4)	Low (due to very serious risk of bias)	0.81	1
Additive substances									
Indigo carmine	2	979	1.08 (0.97 to 1.21)	54.4	59.4	5.0 (–1.2 to 11.2)	Low (due to very serious risk of bias)	0.68	2
Oral methylene blue	1	1002	1.18 (1.04 to 1.33)	47.8	56.4	8.6 (2.4 to 14.7)	Moderate (due to serious imprecision)	0.94	1
Menthol	3	698	0.94 (0.79 to 1.11)	45.9	42.3	–3.6 (–10.3 to 3.0)	Moderate (due to serious risk of bias)	0.05	3

PEG, polyethylene glycol.

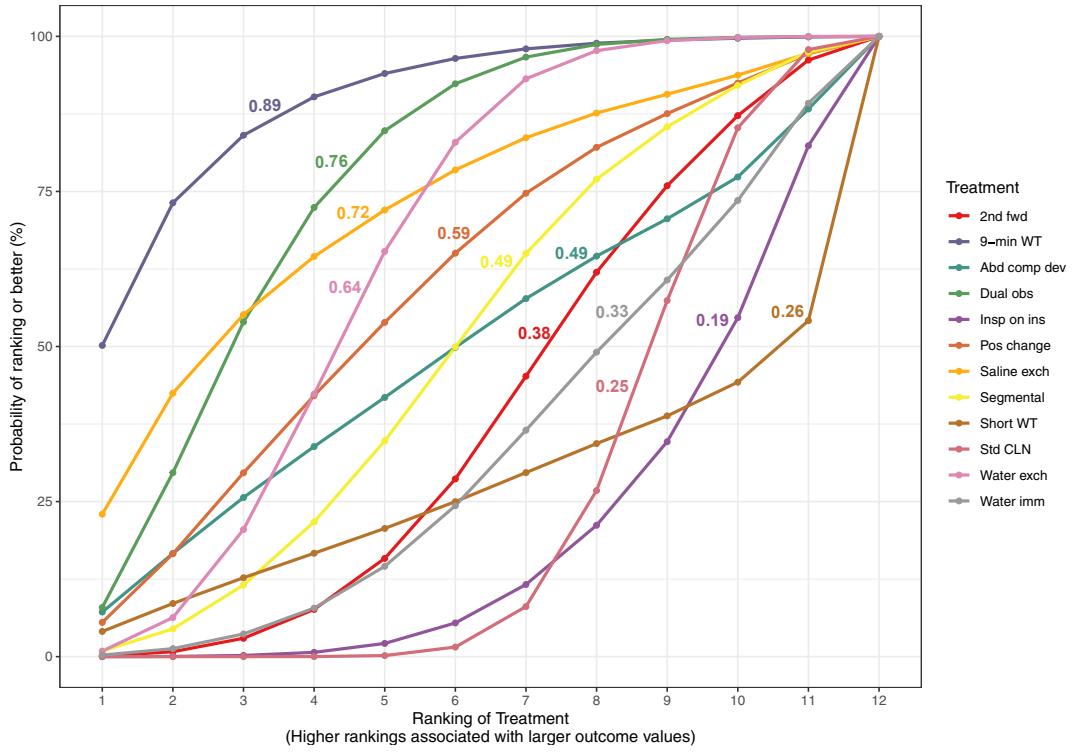


Figure 3. SUCRA curve for intraprocedural techniques. 2nd fwd, second inspection in forward view; 9-min WT, 9-minute withdrawal time; Abd comp dev, abdominal compression device; Dual obs, dual observation; Insp on ins, inspection on insertion; Pos change, dynamic position changes; Saline exch, saline exchange; Segmental, segmental timed withdrawal; Short WT, short withdrawal time; Std CLN, standard colonoscopy; Water exch, water exchange; Water imm, water immersion.

Subgroup Analyses

For the subgroup analysis based on colonoscopy indication, we conducted network meta-analysis for 1 domain (distal attachment devices) for studies with only patients with positive fecal immunochemical test or occult blood test results and 5 domains (periprocedural parameters, intraprocedural techniques, endoscopic technologies, distal attachment devices, additive substances) for studies with colonoscopy as the initial screening modality. There were too few studies in the other domains for the fecal immunochemical test or occult blood test subgroups to conduct network meta-analysis. Among the subgroups with positive fecal immunochemical testing or occult blood testing results, no intervention in the distal attachment devices domain resulted in improved ADR. In the initial screening subgroup, 9-minute withdrawal time (1 trial, 733 participants; OR, 1.26; 95% CrI, 1.05–1.52) and linked color imaging (3 trials, 1917 patients; OR, 1.22; 95% CrI, 1.05–1.43) in the endoscopic technologies domain and first-generation Endocuff (1 trial, 337 patients; OR, 1.74; 95% CrI, 1.06–2.87) in the distal attachment devices domain resulted in improved ADR.

We did not perform subgroup analyses for patient demographic characteristics (age, sex) given the lack of data on ADR stratified by these characteristics in individual studies, for bowel preparation quality and sedation practices due to variable reporting of these variables, for colonoscopy completion rates due to almost 100% completion rates, and for endoscopist specialty and trainee participation due to infrequent reporting of these variables.

Sensitivity Analyses

In the sensitivity analysis only including studies published in 2015 and after, all interventions resulting in improved ADR in the main analysis did so in the sensitivity analysis as well, except for dual observation and i-SCAN. In the sensitivity analysis that only included studies based in North America, Endocuff Visio became the only intervention that resulted in improved ADR. Full results of sensitivity analyses with updated estimates are provided in Supplementary File 4. We did not perform sensitivity analyses for high study attrition because no study had attrition of >25%, for studies without clear descriptions of comparator arms because all studies described control groups in detail, for studies with more restrictive screening-related criteria because there were too few studies with a >95% screening population, or for studies at high risk of bias because the vast majority of studies were at high risk of bias.

Risk of Bias

There were 6 trials at low risk of bias, 13 with some concerns, and 105 at high risk of bias (with a visual summary of the risk of bias provided in Supplementary Figure 2 and detailed risk of bias assessments provided in

Table 4. Secondary Outcomes With Effect Estimates and 95% CrIs

Intervention	PDR, OR (95% CrI)	ssADR, OR (95% CrI)	rADR, OR (95% CrI)	aADR, OR (95% CrI)	APC, MD (95% CrI)	ssAPC, MD (95% CrI)	WT, MD (95% CrI)	CIR, RR (95% CrI)
Periprocedural parameters								
Bisacodyl	1.17 (0.83 to 1.64)	—	—	—	—	—	—	1.00 (0.98 to 1.02)
Coffee plus PEG	1.25 (0.91 to 1.74)	—	—	—	—	—	—	—
Electrolyte-free PEG	0.95 (0.73 to 1.27)	—	—	—	—	—	—	—
Low-dose PEG	1.14 (0.93 to 1.41)	—	1.42 (0.62 to 3.34)	—	—	—	0.60 (−0.78 to 1.99)	—
Low-fiber diet	1.03 (0.85 to 1.21)	—	1.33 (0.75 to 2.38)	—	—	—	—	1.01 (0.95 to 1.06)
Low-fiber diet and sulfate-free PEG	1.09 (0.77 to 1.51)	—	—	—	—	—	—	1.01 (0.98 to 1.04)
Patient education	0.91 (0.77 to 1.07)	—	—	—	—	—	0.40 (−0.65 to 1.40)	—
Simethicone	0.86 (0.65 to 1.11)	—	0.92 (0.51 to 1.69)	—	—	—	−0.04 (−1.08 to 0.99)	—
Single-dose PEG	—	—	—	—	—	—	−0.40 (−1.79 to 0.99)	—
Sulfate-based preparation	0.99 (0.84 to 1.13)	—	0.96 (0.58 to 1.60)	—	—	—	0.20 (−1.22 to 1.63)	1.00 (0.97 to 1.05)
Sulfate-free PEG	1.17 (0.83 to 1.60)	—	—	—	—	—	—	—
Intraprocedural techniques								
9-minute withdrawal time	1.25 (0.88 to 2.29)	—	1.35 (1.06 to 1.74)[a]	1.07 (0.55 to 2.12)	0.21 (–0.20 to 0.62)	0 (–0.25 to 0.25)	2.95 (–3.31 to 5.93)	
Abdominal compression device	0.91 (0.63 to 1.30)	—	0.90 (0.62 to 1.29)	—	0.16 (−0.30 to 0.63)	—	−0.90 (−7.31 to 5.46)	—
Dual observation	1.13 (0.88 to 1.49)	—	1.00 (0.80 to 1.24)	1.19 (0.76 to 1.86)		—	2.24 (−1.37 to 5.93)	1.00 (0.91 to 1.09)
Inspection on insertion	—	—	1.02 (0.81 to 1.30)	1.00 (0.65 to 1.55)	−0.01 (−0.38 to 0.28)	—	−3.04 (−7.62 to 1.43)	—
Prescribed positional change	0.99 (0.70 to 1.42)	—	—	—	—	—	0.80 (−5.48 to 7.07)	—
Proximal colon retroflexion	—	—	0.74 (0.46 to 1.17)	—	—	—	—	—
Saline exchange	—	—	1.15 (0.85 to 1.55)			0 (–0.36 to 0.36)	−3.45 (−9.52 to 2.42)	1.01 (0.95 to 1.08)
Second forward view	1.09 (0.87 to 1.35)	—	0.33 (0.29 to .38)	1.11 (0.70 to 1.78)	—	—	1.01 (−2.63 to 4.68)	—
Segmental timed withdrawal	1.16 (0.79 to 1.72)	—	1.27 (0.80 to 2.03)	0.63 (0.25 to 1.46)	0.14 (−0.33 to 0.61)	—	−0.00 (−6.27 to 6.25)	1.00 (0.93 to 1.08)
Short withdrawal time	—	—	—	—	—	—	—	—
Water exchange	1.33 (0.92 to 1.93)	—	1.15 (1.02 to 1.31)[a]	1.18 (0.92 to 1.51)	–0.10 (–0.85 to 0.66)	–0.20 (–0.53 to 0.13)	−1.05 (–4.31 to 2.06)	1.00 (0.96 to 1.03)
Water immersion	—	—	1.02 (0.79 to 1.31)	1.10 (0.82 to 1.48)	–0.19 (–0.78 to 0.39)	—	0.26 (−4.01 to 4.52)	1.00 (0.94 to 1.05)
Endoscopic technologies								
Blue light imaging	—	—	—	—	—	—	—	—
CADe	1.06 (1.02 to 1.12)[a]	1.07 (0.81 to 1.36)	1.29 (0.86 to 1.93)	1.00 (0.83 to 1.25)	0.16 (0.06 to 0.26)[a]	—	0.27 (−0.02 to 0.62)	1.00 (0.97 to 1.03)
Flexible spectral imaging color enhancement	0.95 (0.84 to 1.08)	—	—	—	−0.01 (−0.10 to 0.08)	—	—	—
i-SCAN	1.03 (0.88 to 1.19)	1.02 (0.55 to 1.91)	—	0.97 (0.52 to 1.91)	0.09 (−0.06 to 0.24)	—	0.25 (−0.18 to 0.77)	—
Linked color imaging	1.14 (1.07 to 1.22)[a]	1.27 (0.63 to 2.50)	1.04 (0.69 to 1.56)	1.05 (0.80 to 1.37)	0.19 (0.09 to 0.30)[a]	—	0.26 (−0.19 to 0.74)	1.00 (0.98 to 1.03)

Table 4. Continued

Intervention	PDR, OR (95% CrI)	ssADR, OR (95% CrI)	rADR, OR (95% CrI)	aADR, OR (95% CrI)	APC, MD (95% CrI)	ssAPC, MD (95% CrI)	WT, MD (95% CrI)	CIR, RR (95% CrI)
Narrow-band imaging	0.99 (0.91 to 1.06)	2.94 (1.46 to 6.25)[a]	1.38 (0.88 to 2.10)	0.92 (0.71 to 1.20)	−0.02 (−0.10 to 0.06)	—	0.61 (0.22 to 1.01)[a]	1.00 (0.99 to 1.01)
Ultrathin colonoscope	0.88 (0.61 to 1.25)	—	0.46 (0.16 to 1.21)	0.26 (0.05 to 0.86)	—	—	—	1.06 (0.99 to 1.15)
Wide-angle colonoscope	0.95 (0.88 to 1.04)	0.92 (0.53 to 1.65)	0.96 (0.52 to 1.76)	—	−0.24 (−0.78 to 0.30)	—	0.26 (−0.33 to 0.87)	0.99 (0.94 to 1.03)
Distal attachment devices								
AmplifEYE	1.09 (0.84 to 1.43)	1.08 (0.57 to 2.08)	—	1.17 (0.59 to 2.27)	0.28 (−0.85 to 1.43)	0.02 (−1.58 to 1.63)	−0.74 (−2.78 to 1.21)	—
Cap-assisted colonoscopy	1.09 (0.89 to 1.33)	0.84 (0.55 to 1.30)	1.03 (0.85 to 1.21)	1.09 (0.83 to 1.46)	0.08 (−0.39 to 0.57)	0.36 (−0.42 to 1.20)	0.12 (−0.87 to 1.05)	1.01 (0.97 to 1.04)
First-generation Endocuff	1.21 (1.06 to 1.42)[a]	1.07 (0.69 to 1.72)	0.99 (0.69 to 1.41)	—	0.78 (−0.31 to 1.88)	—	0.19 (−0.88 to 1.27)	1.01 (0.97 to 1.05)
Endocuff Vision	1.09 (0.97 to 1.23)	1.15 (0.81 to 1.63)	1.12 (0.97 to 1.30)	1.05 (0.84 to 1.28)	0.16 (−0.38 to 0.72)	0.06 (−1.08 to 1.20)	−0.44 (−1.16 to 0.18)	1.01 (0.98 to 1.05)
EndoRings	1.03 (0.90 to 1.18)	1.09 (0.76 to 1.57)	1.23 (0.79 to 1.92)	0.88 (0.55 to 1.39)	0.07 (−0.59 to 0.77)	0.06 (−1.07 to 1.20)	−0.72 (−1.99 to 0.54)	0.99 (0.94 to 1.05)
WingCap	1.21 (0.90 to 1.64)	0.54 (0.13 to 1.87)	—	1.18 (0.46 to 3.06)	0.27 (−0.67 to 1.21)	—	0.36 (−1.29 to 2.03)	1.00 (0.95 to 1.06)
Additive substances								
Indigo carmine	—	—	—	—	—	—	—	—
Oral methylene blue	—	—	—	—	—	—	—	—
Menthol	—	—	—	—	—	—	—	—

aADR, advanced ADR; APC, adenomas per colonoscopy; CIR, cecal intubation rate; MD, mean difference; PDR, polyp detection rate; PEG, polyethylene glycol; rADR, right ADR; RR, risk ratio; ssADR, sessile serrated ADR; ssAPC, sessile serrated adenomas per colonoscopy; WT, withdrawal time.
[a]Significant effect with $P < .05$.

Supplementary Figure 3). Most had low risk of bias related to randomization (112 low risk of bias, 9 some concerns, 3 high risk of bias). Most trials had high risk of bias due to deviation from the intended intervention (7 low risk of bias, 14 some concerns, 103 high risk of bias) and measurement of the outcome (22 low risk of bias, 102 high risk of bias) because study investigators and endoscopists performing the procedures and those assessing outcomes were often unblinded to whether participants received the intervention. Most trials had low risk of bias in the missing outcome domain (122 low risk of bias, 2 some concerns), and all had low risk of bias in the selective reporting domain.

Certainty of the Evidence

Certainty of evidence for the various interventions for all other outcomes ranged from very low to moderate. For the primary outcome of ADR, overall certainty of evidence was very low for 10 of 38 interventions, low for 20 interventions, and moderate for 7 interventions (Table 3). Certainty in the strength of evidence for individual studies was downgraded for risk of bias (6 interventions serious, 25 very serious), inconsistency (5 serious), and imprecision (15 serious). There were no interventions with serious incoherence or indirectness because most comparisons were only based on direct evidence.

Discussion

We assessed a broad range of interventions compared to standard colonoscopy within a homogeneous screening-related population in this network meta-analysis. Nine of 37 interventions across multiple domains resulted in significant improvements in ADR, and others ranked highly in SUCRA analyses, indicating their overall potential for ADR improvement as well. There are several key findings. First, 9-minute withdrawal, water exchange, and dual observation—all techniques that can be used without purchasing additional equipment and are therefore immediately and widely implementable—resulted in improved ADR. Similarly, within the endoscopic technologies domain, linked color imaging and i-SCAN improved ADR, and narrow-band imaging improved detection of serrated adenomas. CADe also improved ADR. Use of several distal attachment devices, including first-generation Endocuff and Endocuff Vision, resulted in improved ADR with no significant adverse effects on cecal intubation time.

The successful identification of adenomas is dependent on both mucosal exposure and lesion recognition. Prior evidence syntheses characterizing the effects of interventions to optimize adenoma detection[12–15,26,27] have not considered ADR within these related but distinct strategies. For example, recent analyses compared distal attachment devices with endoscopic technologies such as CADe.[13,15] Comparing these 2 kinds of interventions may be problematic because distal attachment devices aim to improve mucosal exposure, whereas endoscopic imaging–based technologies aim to improve recognition of lesions that are already onscreen. We addressed this gap by comparing interventions within predefined domains, which we believe more accurately reflects clinical practice, where interventions can be readily combined (ie, using a distal attachment device and virtual chromoendoscopy) and where cost considerations are likely more comparable (ie, double right-sided observation and dual observation are not associated with increased direct costs, whereas distal attachment devices are). Using our results, endoscopists, unit managers, and policy makers can identify the most effective interventions within each domain.

Despite several interventions demonstrating significant efficacies in improving ADR detection, it is noteworthy that none improved detection of advanced adenomas, lesions associated with significantly higher CRC risk.[28] Therefore, the primary reason behind ADR improvements we observed is detection of smaller or less histologically aggressive adenomas, whose risks of progression to CRC are considerably lower.[29] Similarly, only the application of narrow-band imaging improved detection of serrated polyps, with no other interventions proving beneficial. These lesions are harder to detect as a result of their flatter morphologic appearance, subtler pit patterns, and potentially indistinct borders.[30] However, their progression to CRC is also more aggressive,[31] making their early detection crucial. Overall, then, one could reasonably question the true value of most interventions we present, given that the detection of arguably the highest-risk lesions is not being enhanced. The findings could be partially because advanced adenomas are easier to detect with standard colonoscopy given their size and/or appearance, and therefore, their (already near-optimal) rate of detection is not meaningfully enhanced by the interventions studied. It could also be that our study was underpowered to detect any potential beneficial effects of individual interventions on advanced ADR given the relative scarcity of patients with large or histologically aggressive polyps. Regardless, future interventions to improve adenoma detection should focus on the detection of advanced adenomas and serrated lesions.

In concordance with our findings, Facciorusso et al[15] found that water exchange and dual observation were associated with improved ADR, and Aziz et al[13] found that first-generation Endocuff, Endocuff Vision, linked color imaging, and CADe were associated with improved ADR. Several of our results, however, are discordant with the existing literature. For example, Facciorusso et al[15] did not find a significant improvement in ADR for WingCap given the publication of a newer study supporting this interventions since publication of their meta-analysis, which was incorporated into our study.[32] This is important given that WingCap ranked highest in our SUCRA analyses, which, together with our other findings, provides more compelling data to support the use of distal attachment devices in screening-related colonoscopy. Another important difference in our approach was our strict eligibility criteria, which created a true screening-based population.[20] In part due to these differences, we found a more modest effect estimate for CADe compared to other evidence syntheses.[32,33] However, similar to existing meta-analyses, we also found no effect of CADe on advanced adenoma or sessile serrated lesion detection, as discussed.[32,33] These results, especially

when viewed within the context of large observational studies that fail to demonstrate significant ADR improvements,[33,34] warrant careful consideration of whether CADe should be adopted in place of imaging modalities that are already built into existing platforms.

Ultimately, the decisions regarding which intervention(s) to adopt at the endoscopist or unit level are nuanced, complex, and highly dependent on practice setting. Factors that should be considered include cost, availability, and feasibility of implementation, among others. One approach is to consider this issue from the standpoint of an economist and create cost-performance ratios for each intervention. This approach, however, is complex and requires careful consideration of capital (one-time) costs vs per-case costs and direct vs indirect costs. For instance, consider that the cost of an arbitrary CADe platform might be $25,000, and the benefit received would be an 11% relative increase in ADR. How does one compare this benefit to that which is afforded by a distal attachment device that costs only $20 and provides a 15% relative ADR increase? At first glance, the latter intervention might be more appealing. A more appropriate comparison, however, would require consideration of factors such as room numbers; annual case volumes; and costs related to repair, maintenance, and/or upgrading of technology. There is also emerging research assessing combinations of interventions that could have even greater benefits than any one intervention alone.[35] More well-performed cost-effectiveness analyses need to be performed to provide sensible estimates on which endoscopists and unit managers can base their decisions and on which learned societies can base their guidance.[36]

Perhaps our most important finding is that we identified several zero-cost interventions that each demonstrated benefits in improving ADR, and most of these can relatively easily be applied immediately by most endoscopists. Three such interventions include 9-minute withdrawal time, dual observation, and water exchange. None of these interventions requires any additional training, nor do they require any investment or result in any major unforeseen indirect costs. A 9-minute withdrawal resulted in a significant increase in ADR based on a single trial.[37] Currently, US-based multisociety guidance recommends a minimum 6-minute withdrawal,[7] whereas European guidance suggests a 10-minute aspirational target with a 6-minute minimum.[38] These data would support this aspirational target. Dual observation could be variably implementable depending on practice settings and room setups, but simple knowledge of these findings should prompt many endoscopists to change their practices today and, as examples, involve themselves to a greater degree when supervising trainees or recruit nursing staff more actively into being active observers when feasible. Water exchange is also a relatively easy-to-adopt technique that, in addition to increasing ADR, has also been associated with higher cleanliness scores compared to standard colonoscopy.[39] In addition to these zero-cost interventions, we found that in the endoscopic technologies domain, both i-SCAN and linked color imaging demonstrated significant improvements in ADR overall, while narrow-band imaging demonstrated significant improvements in detection of sessile serrated lesions. Therefore, another key message we have uncovered is that regardless of the endoscopy platform an endoscopist has available to them, there exist dedicated modalities that can enhance adenoma detection. Endoscopists performing screening-related colonoscopy should be encouraged to contact their endoscope manufacturer representatives to learn more about these technologies and how to apply them, given that they are associated with no additional costs.

Our study has several strengths. As mentioned, prior network meta-analyses reported results across multiple interventional domains and, crucially, included patients undergoing colonoscopy for non-screening-related indications.[12–15,26,27] These factors can introduce significant potential indirectness when applying these interventions to one's screening colonoscopy practice. Unsurprisingly, these studies reported substantial statistical heterogeneity and undetected heterogeneity. Compared to prior studies, we found substantial statistical heterogeneity for only 4 interventions. We also assessed the transitivity assumption (the assumption of a lack of important imbalances between studies included in the network), a critical component of network meta-analysis,[16] which was inconsistently tested in prior syntheses. Given that differences in study populations and associated baseline risk can violate transitivity, this is especially critical for our research question. Furthermore, prior evidence syntheses did not assess the certainty of evidence using the GRADE framework,[25] which is a major gap. Finally, we published our study protocol a priori to promote transparency and enhance the reproducibility of our analysis.

Our study also has several limitations. Importantly, we excluded observational studies. Although this approach improved the overall certainty of the evidence, we may have missed novel and/or emerging interventions not yet studied in a trial. Furthermore, although only randomized trials can provide data on efficacy, they can be prone to issues regarding generalizability.[40] Prior studies assessing screening colonoscopy interventions and colonoscopic interventions in general have demonstrated differences between efficacy and real-world effectiveness.[33,41] Second, we excluded conference abstracts. Although this may have introduced publication bias, we found no statistical evidence for this in our analyses. Third, although we assessed transitivity using patient age and sex as potential effect modifiers, we were unable to include other variables, such as bowel preparation quality, given the lack of meaningful variation in bowel preparation quality among studies (ie, >90% of included procedures had adequate or better bowel cleansing). Finally, we were not able to meaningfully analyze or present data on the impact(s) of any intervention(s) on outcomes such as adverse event rates or postcolonoscopy CRC. Although the primary and secondary outcomes we present are well established surrogates for postcolonoscopy CRC, we are not aware of any existing randomized trials that assess the impacts of such interventions directly on postcolonoscopy CRC.

Our results provide important information regarding the relative efficacies of interventions to improve colonoscopy

quality. Importantly, we uncovered 3 efficacious intraprocedural techniques (9-minute withdrawal, dual observation, and water exchange) and an additional 3 technologies from individual endoscope manufacturers that also boost adenoma detection (linked color imaging and i-SCAN) or sessile serrated adenoma detection (narrow-band imaging), all 6 with no associated additional costs. These results are relevant to endoscopy unit managers, gastrointestinal and endoscopy societies that produce clinical guidelines, and practicing endoscopists. Future original studies should begin to assess the efficacy and real-world effectiveness of practical combinations of these interventions applied together. Future research should also use these high-quality data to perform cost-effectiveness analyses of interventions within each domain.

Supplementary Material

Note: To access the supplementary material accompanying this article, visit the online version of *Gastroenterology* at www.gastrojournal.org, and at https://doi.org/10.1053/j.gastro.2024.03.018.

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Data Availability
All data, analytic methods, and study materials can potentially be made available directly from the corresponding author upon review following reasonable request.